

Disclaimer:

This presentation is confidential and may not be distributed, transmitted or otherwise communicated, in whole or in part, without the express written consent of Numerai. This presentation does not constitute an offer to sell, or the solicitation of an offer to buy, any security, product or service, including an interest in any fund. Unless otherwise noted, information included herein is presented as of the dates indicated. This presentation is not complete and the information contained herein may change at any time without notice. Numerai does not have any responsibility to update the presentation to account for such changes. Numerai makes no representation or warranty, expressed or implied, with respect to the accuracy, reasonableness, or completeness of any of the information contained herein, including, but not limited to, information obtained from third parties. It should not be assumed that performance of investments made in the future will equal or have any correlation to the investments discussed in the attached materials and it should not be assumed that opportunities in the future will be profitable. The information contained herein is not intended to provide, and should not be relied upon for, accounting, legal or tax advice or investment recommendations. Investors should make an independent investigation of the investment described herein, including consulting their tax, legal, accounting or other advisors, about the matters, discussed herein. Past performance is not indicative, nor a guarantee of, future results. Losses may occur. Certain information contained herein may be "forward-looking" in nature. Because these forward-looking statements are based on estimates and assumptions that are subject to significant business, economic and competitive uncertainties, many of which are beyond Numerai's control or are subject to change, actual results could be materially different. Due to various risks and uncertainties, actual events or results or the actual performance may differ materially from those reflected or contemplated in such forward-looking information. As such, undue reliance should not be placed on such information. Forward-looking statements may be identified by the use of terminology including, but not limited to, "may," "will," "should," "expect," "anticipate," "target," "project," "estimate," "intend," "continue" or "believe" or the negatives thereof or other variations thereon or comparable terminology. Numerai cannot give any assurance that the assumptions upon which it based its forward-looking statements will prove to be correct. There can be no assurance that the investment objectives described herein will be achieved. Nothing herein is intended to imply that an investment methodology may be considered "conservative," "safe," "risk free," or "risk averse." Economic, market and other conditions could also cause to alter its investment objectives, guidelines and restrictions. Investment losses may occur. Additional information may be available upon request.



NUMERAI

Michael Oliver
Chief Scientist @ Numerai

Legal Disclaimer

This presentation is not investment advice and is solely for informational purposes. It does not constitute an offer or solicitation of an offer or any advice or recommendation to purchase any securities or other financial instruments and may not be construed as such.

Past performance is no guarantee of future results.

<https://numer.ai/terms>

What is Numerai?

**A market neutral, global equities
hedge fund.**

**The hardest data science
tournament on the planet.**

**Also the highest paying.
\$40M+ paid out.**

Who is Michael? Chief scientist at
Numerai

Formerly a computational
neuroscientist

What is a neuroscientist doing at a
hedge fund?

Neuroscience data is ...

- Low signal-to-noise
- High dimensional
- Non-linear
- Non-stationary
- Under-sampled

Finance data is ...

- Low signal-to-noise
- High dimensional
- Non-linear
- Non-stationary
- Under-sampled



NUMERAI

There are many smart people out
there doing amazing things

Numerai unleashes their intelligence
on the stock market

id	era	feature1	feature2	target
n2b2e3dd163cb422	era1	0.25	1.00	0.50
n177021a571c94c8	era1	0.75	0.00	0.50
n7830fa4c0cd8466	era1	0.50	0.50	1.00
nc584a184cee941b	era1	0.25	1.00	0.00
nc5ab8667901946a	era1	1.00	0.50	0.75
n84e624e4714a7ca	era1	0.50	0.25	1.00
n6c3f4ad216ac213	era1	0.00	0.70	0.50

Hedge fund quality data
cleaned, obfuscated,
and **given away** to be
modeled

Train once,
Predict daily

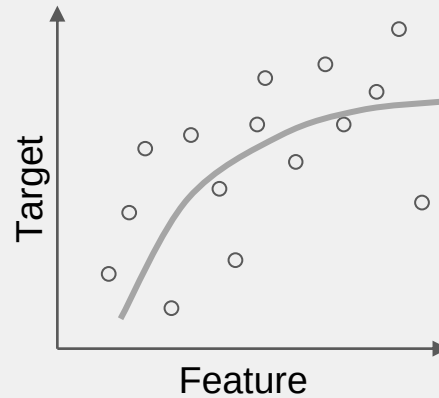
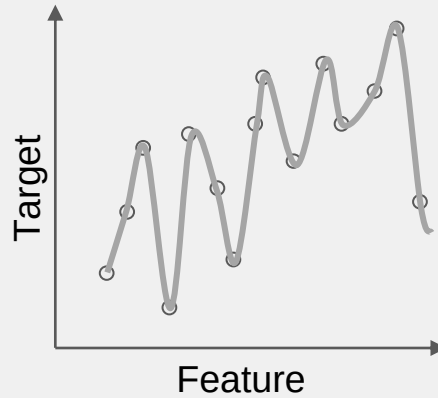
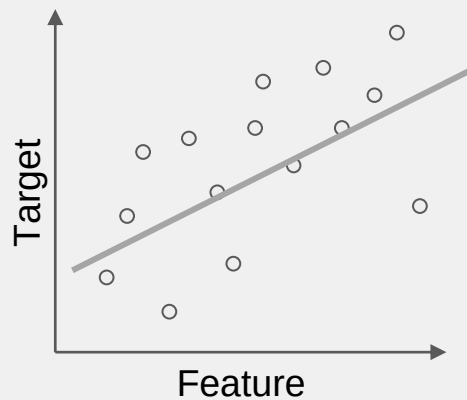
Stake on your
predictions with **NMR**

Earn NMR / Burn NMR

Financial data is way smaller than it seems

- We don't just care if AAPL went up
- We care if AAPL went up and everything else went down
- 1 era = 1 sample
- So we have 1000 samples, and 2000 features
- So these dimensions are... terrible for ML

The Art of Machine Learning



Staking

- "Skin in the game" lets us know if a submission is serious + high confidence
- Lets us know how much to weight each user
- Self-correcting feedback mechanism
- Over \$4.5 million worth of NMR staked

Model Upload for easy automation



```
# Define your prediction pipeline as a function
```

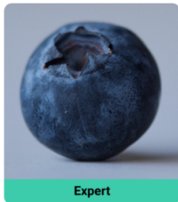
```
def predict(live_features: pd.DataFrame) -> pd.DataFrame:  
    live_predictions = model.predict(live_features[feature_cols])  
    submission = pd.Series(live_predictions, index=live_features.index)  
    return submission.to_frame("prediction")
```



```
# Use the cloudpickle library to serialize your function
```

```
import cloudpickle  
p = cloudpickle.dumps(predict)  
with open("predict.pkl", "wb") as f:  
    f.write(p)
```

We also provide tools for self-hosting models



Expert

BLUEBERRY MUFFIN

Joined 4 years ago

A non-random walk down Wall Street.

MMC Rank

6th

of 2,173

CORR Rank

201st

of 2,173

Season Rank

27th

of 395

Stake

10,341 NMR

\$88,000

1Y Return

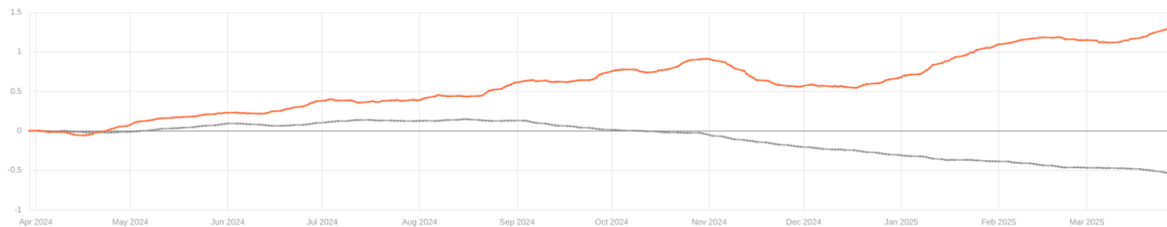
84%

\$33,270.04

MMC

☒ Numeral Benchmark Models 

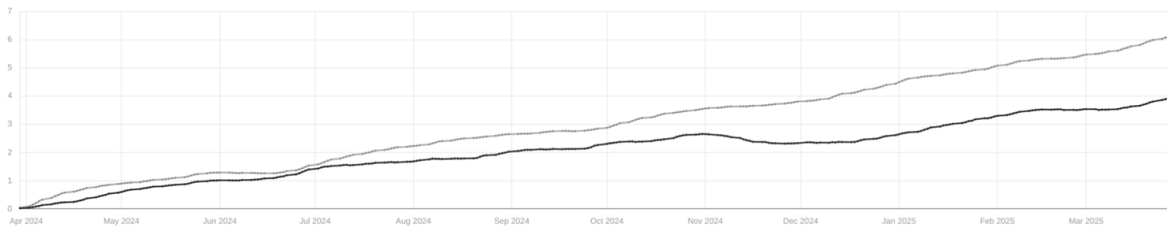
Cumulative stake-weighted-average MMC score of all models over the past 1 year vs Numeral benchmark models



CORR

☒ Numeral Benchmark Models 

Cumulative stake-weighted-average CORR score of all models over the past 1 year vs Numeral benchmark models



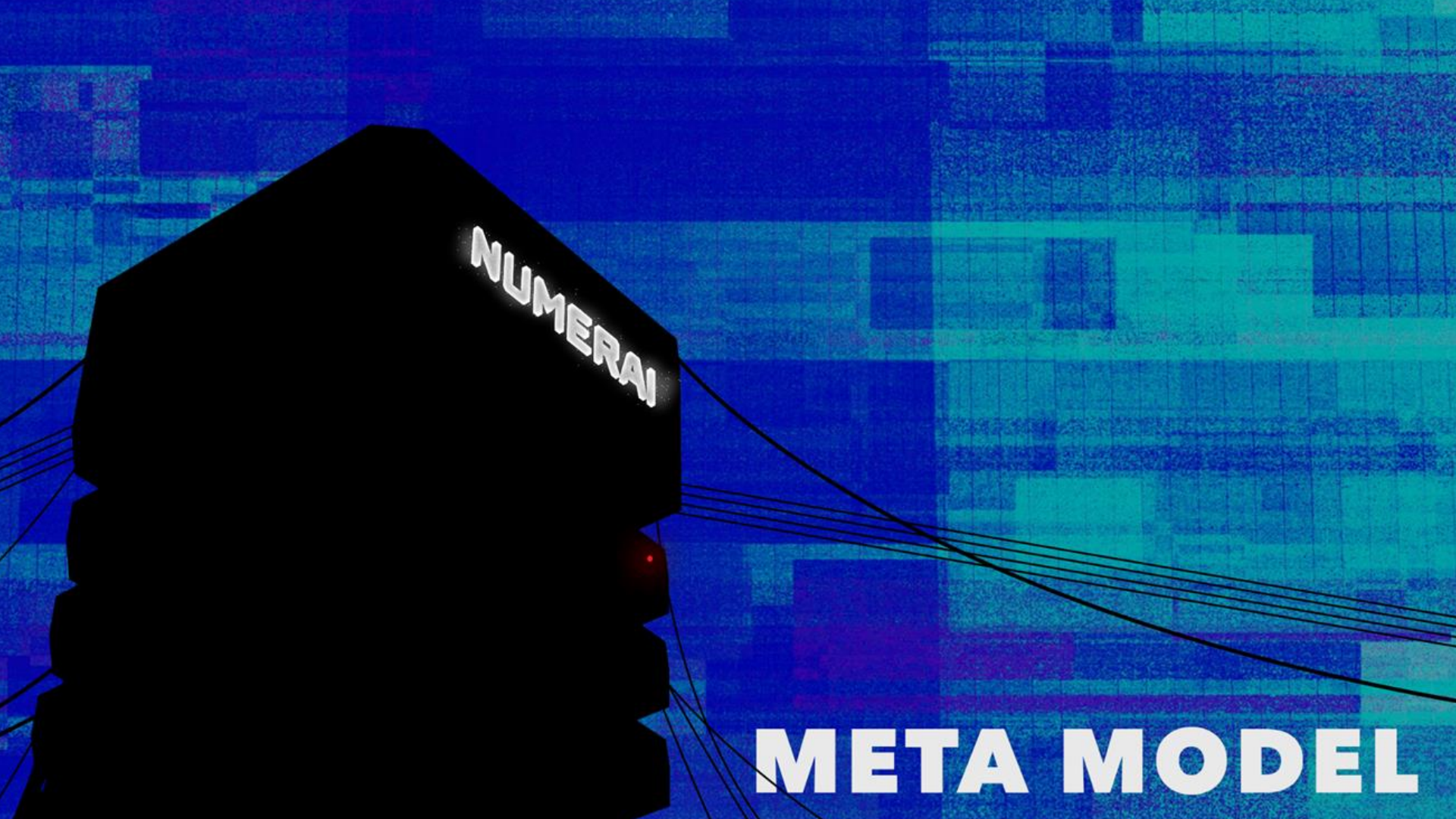
Leaderboard shows
current reputation and
stake values

Scores change daily with
market movements

Payouts happen daily
when 4-week long round
resolves

Predictions from each
user are aggregated,
weighted by stake to
form ...

Data Scientists		Models									
		Staked Models		Staked NMR At Risk				Avg 1Y Return			
		3,990		585,811 NMR				47.65%			
RANK	NAME	COR360	FNCV3	CORR20V2	• MMC	BMC	CORT20	STAKE	1D	3M	1Y
1	 GAMMARAT_076	0.0193	0.0163	0.0163	0.0103	0.0115	0.0101	—	—	—	—
2	 GAMMARAT_083	0.0188	0.0129	0.0152	0.0098	0.0108	0.0113	—	—	—	2.5%
3	 DORA_08	0.0221	0.0184	0.0185	0.0095	0.0121	0.0153	0.026 NMR	0.10%	38.6%	149.2%
4	 FORS002	0.0444	0.0208	0.0230	0.0093	0.0121	0.0207	7.273 NMR	0.96%	47.7%	—
5	 PL11_STANISLAW_LEM	0.0353	0.0228	0.0229	0.0093	0.0125	0.0139	1.137 NMR	2.6%	40.9%	177.4%
6	 GAMMARAT_054	0.0267	0.0120	0.0134	0.0089	0.0100	0.0092	—	—	—	—
7	 GAMMARAT22	0.0172	0.0108	0.0126	0.0089	0.0098	0.0084	—	—	—	22.1%
8	 GAMMARAT36	0.0264	0.0131	0.0146	0.0087	0.0099	0.0103	2 NMR	1.8%	—	—
9	 GAMMARAT_080	0.0157	0.0134	0.0150	0.0085	0.0096	0.0120	—	—	—	76.2%
10	 GAMMARAT33	0.0245	0.0099	0.0142	0.0085	0.0098	0.0110	—	—	—	67.3%
11	 GAMMARAT14	0.0222	0.0144	0.0152	0.0085	0.0097	0.0100	2 NMR	0.10%	—	—
12	 PL8_STEFAN_BANACH	0.0381	0.0196	0.0226	0.0084	0.0119	0.0158	0.086 NMR	3.0%	39.6%	150.5%
13	 MDEL0022	0.0118	-0.0027	0.0094	0.0084	0.0089	0.0012	—	—	—	—
14	 GLAS007	0.0488	0.0213	0.0237	0.0084	0.0113	0.0231	6.826 NMR	2.6%	38.7%	—
15	 GAMMARAT05	0.0143	0.0138	0.0148	0.0083	0.0096	0.0100	—	—	—	—
16	 GAMMARAT_092	0.0168	0.0101	0.0127	0.0082	0.0092	0.0117	2 NMR	1.1%	—	—
17	 GAMMARAT_099	0.0176	0.0116	0.0138	0.0081	0.0090	0.0079	—	—	—	—
18	 GAMMARAT_067	0.0147	0.0124	0.0129	0.0079	0.0089	0.0076	—	—	—	—



NUMERAIR

META MODEL

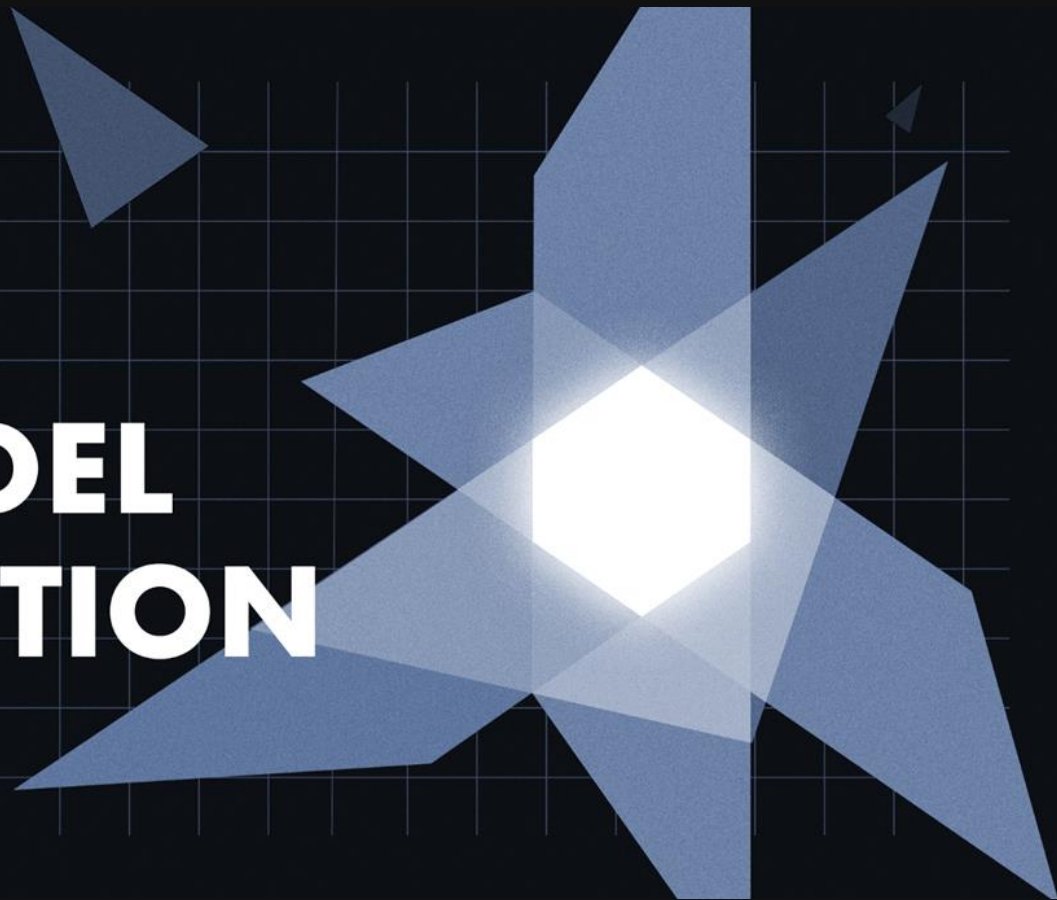
Diversity is King

- "Diversity" in other ML problems isn't very high priority
- For us, it is maybe the highest priority
- Okay performance and high uniqueness is more valuable than high performance but not much different than something we already know
- The market itself is an ensemble of diverse opinions
- So we think the Meta Model is the best way to predict it

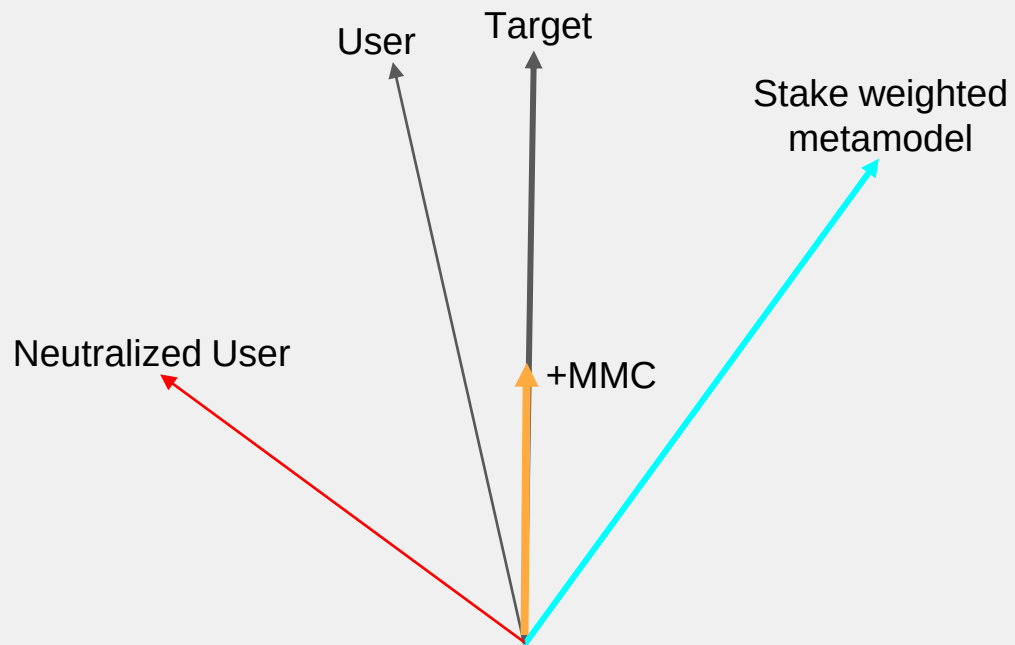


NUMERA I

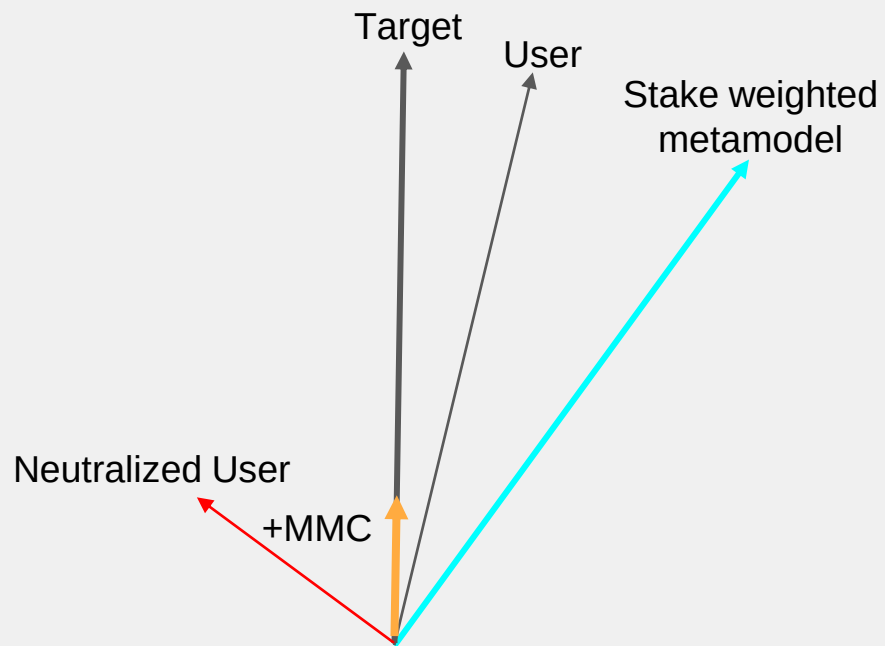
META MODEL CONTRIBUTION



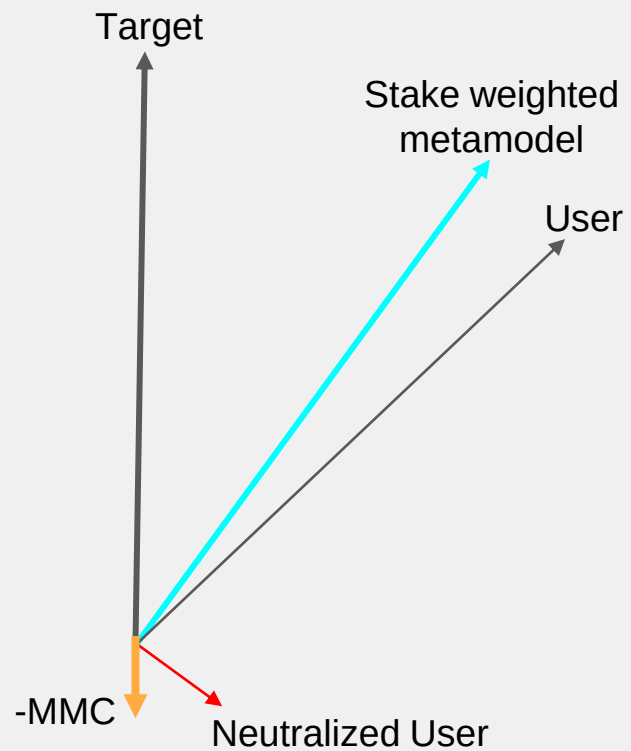
MMC



MMC



MMC



Incentive alignment

Users want to make money – We want to make money

Users want to contribute to the Meta Model prediction of targets

Targets must be amenable to modeling and reflect stock returns

Higher correlation to targets : more profitable portfolios

Lower correlation to targets : less profitable portfolios

Goodhart's law:

"When a measure becomes a target, it ceases to be a good measure"

Why this is hard

Churn – signals that change too quickly can't be traded profitably

Liquidity & market impact – we can't buy full positions in every stock

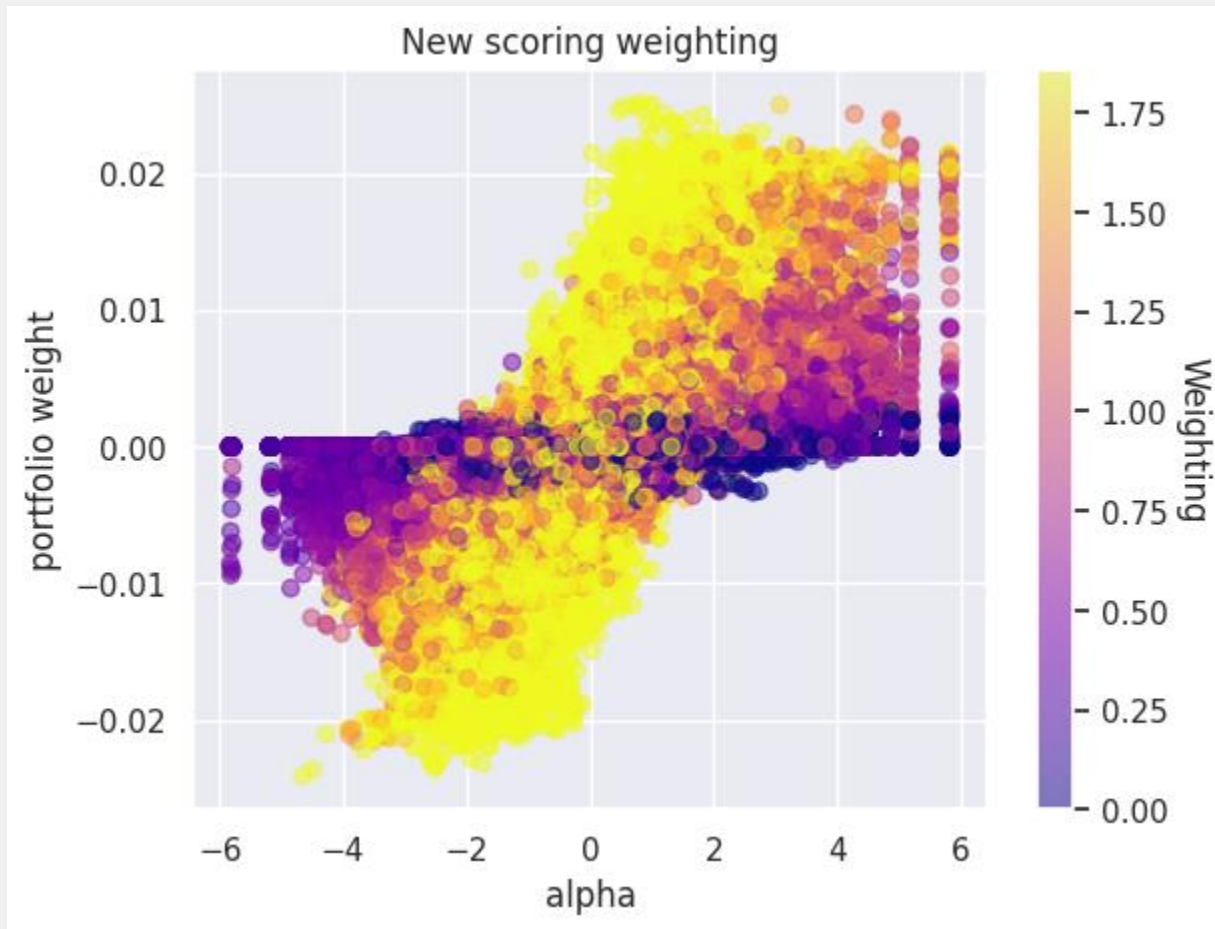
Volatility & risk – we don't necessarily want full positions in every stock

Why this is hard

These constraints can make signals that have the same correlation to stock returns generate portfolios with very different profitability

Very difficult to incorporate all these constraints into targets that still reflect stock returns and are amenable to modeling

Better sample weighting coming soon



Thanks



Come visit us: [HTTP://NUMER.AI](http://numer.ai)